

Enhancing Digital Government Economy (EDGE)

Digital Skills for Students

Host University: Chittagong University of Engineering and
Technology

Host Dept./Institute/Center: Dept. of CSE, CUET

Venue: Southeast University,

Batch Number: CADS-003

Training Track/Course Name: Applied Penetration Testing
Engineer

Hands-on Level: Advanced Level

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Student ID: 2024100000465

Scenario: 01

Executing “tryhackme.com” and using “theHarvester -d tryhackme.com -b all” to find 5 unique subdomains.

Founded Domains are:

Subdomain	IP Address	Notes
setup.auth.tryhackme.com	104.18.35.233	Authentication service
squawker-vpn.vm.tryhackme.com	18.164.78.9	VPN service
status-api.vm.tryhackme.com	18.200.80.242	API endpoint
status.vm.tryhackme.com	3.171.139.54	Status service
store.tryhackme.com	23.227.38.74	E-commerce
store.tryhackme.com	shops.myshopify.com	Third-party hosting

Screenshot of Founded Domains:

```
setup.auth.tryhackme.com
setup.auth.tryhackme.com:104.18.35.233
squawker-vpn.vm.tryhackme.com
squawker-vpn.vm.tryhackme.com:18.164.78.9
status-api.vm.tryhackme.com
status-api.vm.tryhackme.com:18.200.80.242
status.vm.tryhackme.com
status.vm.tryhackme.com:3.171.139.54
store.tryhackme.com
store.tryhackme.com:23.227.38.74
store.tryhackme.com:shops.myshopify.com
vm.tryhackme.com
za.tryhackme.com
```

Screenshot of Command:

[illegible]

Finding the DNS report using “nslookup -type=mx tryhackme.com” for mail server & “nslookup -type=ns tryhackme.com” for name server.

Founded Mail Server's Are:

Record Type	Value	Description
MX	alt1.aspmx.l.google.com	Google mail server
MX	alt2.aspmx.l.google.com	Backup mail server
MX	alt3.aspmx.l.google.com	Backup mail server
MX	alt4.aspmx.l.google.com	Backup mail server

Screenshot of Mail Server's:

```
(root@kali)-[/home/kali/Desktop]
# nslookup -type=mx tryhackme.com
Server:          192.168.0.1
Address:         192.168.0.1#53

Non-authoritative answer:
tryhackme.com    mail exchanger = 5 alt1.aspmx.l.google.com.
tryhackme.com    mail exchanger = 10 alt4.aspmx.l.google.com.
tryhackme.com    mail exchanger = 1 aspmx.l.google.com.
tryhackme.com    mail exchanger = 5 alt2.aspmx.l.google.com.
tryhackme.com    mail exchanger = 10 alt3.aspmx.l.google.com.

Authoritative answers can be found from:
```

Founded Name Servers Are:

Record Type	Value	Description
NS	uma.ns.cloudflare.com	Primary DNS server
NS	kip.ns.cloudflare.com	Secondary DNS server

Screenshot of Name Server's:

```
(root@kali)-[/home/kali/Desktop]
# nslookup -type=ns tryhackme.com
Server:          192.168.0.1
Address:         192.168.0.1#53

Non-authoritative answer:
tryhackme.com    nameserver = uma.ns.cloudflare.com.
tryhackme.com    nameserver = kip.ns.cloudflare.com.

Authoritative answers can be found from:
```

Founded DNS record using “whatweb tryhackme.com”:

Component	Value	Description
Web Server	Cloudflare	Reverse proxy & protection
IP Address	172.66.164.239	Public server IP
Technologies	HTML5, JavaScript	Web technologies used
Title	TryHackMe Platform	Website identity

Screenshot of DNS Report:

```
(root@kali)-[/home/kali/Desktop]
# whatweb tryhackme.com
http://tryhackme.com [301 Moved Permanently] Country[RESERVED][ZZ], HTTPServer[cloudflare], IP[172.66.164.239], RedirectLocation[https://tryhackme.com/], UncommonHeaders[cf-ray]
https://tryhackme.com [200 OK] Cookies[thm-aid], Country[UNITED STATES][US], Email[abkhdcbnkasdbncaksd@gg.com,adil.harti@tryhackme.com,adileharti94@gmail.com,ahmad@tryhackme.com,aicnaksdn@gg.com,alex1122@test.com,amine20solo20@gmail.com,dbkbkjdsck222@gg.com,djando1122@test.com,doktur@tryhackme.com,duarte.cruz+1421456@tryhackme.com,free-marketing@tryhackme.com,garvit.pareek@gmail.com,garvit.pareek@tryhackme.com,garvitfreetrial@test.com,georgecosmintaschina1631@gmail.com,ioanna.kyprianou+free@tryhackme.com,ioanna.kyprianou@tryhackme.com,isac+FreeProd@tryhackme.com,jonathan.martins@tryhackme.com,larissa@tryhackme.com,liza.vul+9418@tryhackme.com,liza.vul@tryhackme.com,manu123@test.com,manutest2233@gmail.com,marcostest112233@gmail.com,monthly-marketing@tryhackme.com,monthlyfs@test.com,multicoupon-marketing@tryhackme.com,multicoupon@test.com,rita.pereira@tryhackme.com,ritasdbpwewereirassssaddsdai234@gmail.com,support@tryhackme.com,testdruvey@test.com,testfreenew@testgar.com,testingmayfs@test.com,testmay2@test.com,yearly-marketing@tryhackme.com], HTML5, HTTPServer[cloudflare], IP[104.20.29.66], Open-Graph-Protocol[website], Script[application/json,application/ld+json,text/javascript], Title[TryHackMe | TryHackMe | Cyber Security Training], UncommonHeaders[cf-ray,cf-cache-status,x-amz-id-2,x-amz-request-id]
https://tryhackme.com/ [200 OK] Cookies[thm-aid], Country[RESERVED][ZZ], Email[abkhdcbnkasdbncaksd@gg.com,adil.harti@tryhackme.com,adileharti94@gmail.com,ahmad@tryhackme.com,aicnaksdn@gg.com,alex1122@test.com,amine20solo20@gmail.com,dbkbkjdsck222@gg.com,djando1122@test.com,doktur@tryhackme.com,duarte.cruz+1421456@tryhackme.com,free-marketing@tryhackme.com,garvit.pareek@gmail.com,garvit.pareek@tryhackme.com,garvitfreetrial@test.com,georgecosmintaschina1631@gmail.com,ioanna.kyprianou+free@tryhackme.com,ioanna.kyprianou@tryhackme.com,isac+FreeProd@tryhackme.com,jonathan.martins@tryhackme.com,larissa@tryhackme.com,liza.vul+9418@tryhackme.com,liza.vul@tryhackme.com,manu123@test.com,manutest2233@gmail.com,marcostest112233@gmail.com,monthly-marketing@tryhackme.com,monthlyfs@test.com,multicoupon-marketing@tryhackme.com,multicoupon@test.com,rita.pereira@tryhackme.com,ritasdbpwewereirassssaddsdai234@gmail.com,support@tryhackme.com,testdruvey@test.com,testfreenew@testgar.com,testingmayfs@test.com,testmay2@test.com,yearly-marketing@tryhackme.com], HTML5, HTTPServer[cloudflare], IP[172.66.164.239], Open-Graph-Protocol[website], Script[application/json,application/ld+json,text/javascript], Title[TryHackMe | TryHackMe | Cyber Security Training], UncommonHeaders[cf-ray,cf-cache-status,x-amz-id-2,x-amz-request-id]
```

Scenario: 2

Conducting a deep technical scan to identify potential entry points on tryhackme.com by doing Advance Scan using Nessus.

Screenshots of Vulnerability Assessment report by Nessus.

Hosts1

Vulnerabilities18

History1

MEDIUM

TLS Version 1.1 Deprecated Protocol

Description

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 lacks support for current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1

As of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher will no longer function properly with major web browsers and major vendors.

Solution

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

See Also

<https://datatracker.ietf.org/doc/html/rfc8996>
<http://www.nessus.org/u?c8ae820d>

Output

TLSv1.1 is enabled and the server supports at least one cipher.

To see debug logs, please visit individual host

Port

Hosts

443 / tcp / wwwtryhackme.com

Hosts1

Vulnerabilities18

History1

MEDIUM

TLS Version 1.0 Protocol Detection

Description

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 has a number of cryptographic design flaws. Modern implementations of TLS 1.0 mitigate these problems, but newer versions of TLS like 1.2 and 1.3 are designed against these flaws and should be used whenever possible.

As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher will no longer function properly with major web browsers and major vendors.

PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2018, except for POS POI terminals (and the SSL/TLS termination points to which they connect) that can be verified as not being susceptible to any known exploits.

Solution

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

See Also

<https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

Output

```
TLSv1 is enabled and the server supports at least one cipher.
```

To see debug logs, please visit individual host

Port

Hosts

Hosts1

Vulnerabilities18

History1

Search Vulnerabilities 🔍 4 Vulnerabilities

☐	Sev ▾	CVSS ▾	VPR ▾	EPSS ▾	Name ▲	Family ▲	Count ▾	⚙
☐	MEDIUM	6.5			TLS Version 1.0 Protocol Detect...	Service detection	6	🕒 ✎
☐	MEDIUM	6.5			TLS Version 1.1 Deprecated Pr...	Service detection	6	🕒 ✎
☐	INFO				TLS Version 1.1 Protocol Detect...	Service detection	6	🕒 ✎
☐	INFO				TLS Version 1.2 Protocol Detect...	Service detection	6	🕒 ✎

Hosts1

Vulnerabilities18

History1

HIGHSSL Medium Strength Cipher Suites Supported (SWEET32)>

Description

The remote host supports the use of SSL ciphers that offer medium strength encryption. Nessus regards medium strength as any encryption that uses key lengths at least 64 bits and less than 112 bits, or else that uses the 3DES encryption suite.

Note that it is considerably easier to circumvent medium strength encryption if the attacker is on the same physical network.

Solution

Reconfigure the affected application if possible to avoid use of medium strength ciphers.

See Also

<http://www.nessus.org/u?df5555f5>
<https://sweet32.info>

Output

```
Medium Strength Ciphers (> 64-bit and < 112-bit key, or 3DES)

Name                      Code      KEX      Auth      Encryption      MAC
-----
DES-CBC3-SHA              0x00, 0x0A  RSA      RSA      3DES-CBC (168)  SHA1

The fields above are :

{Tenable ciphername}
```

Hosts1

Vulnerabilities18

History1

Search Vulnerabilities

4 Vulnerabilities

☐	Sev ▾	CVSS ▾	VPR ▾	EPSS ▾	Name ▲	Family ▲	Count ▾		
☐	HIGH	7.5	6.1	0.3775	SSL Medium Strength Cipher S...	General	6	🕒	✎
☐	INFO				SSL Cipher Block Chaining Ciph...	General	6	🕒	✎
☐	INFO				SSL Cipher Suites Supported	General	6	🕒	✎
☐	INFO				SSL Perfect Forward Secrecy Ci...	General	6	🕒	✎

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Configure

Audit Trail

Hosts 1

Vulnerabilities 18

History 1

Filter

Search Vulnerabilities

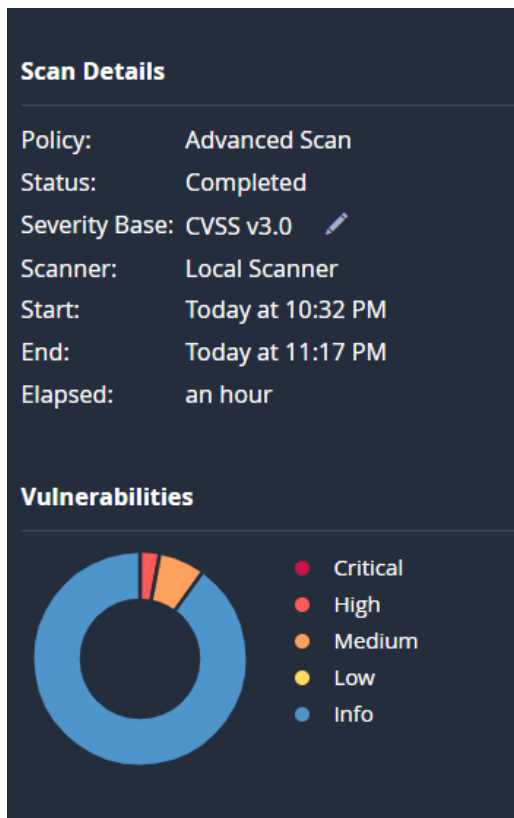
18 Vulnerabilities

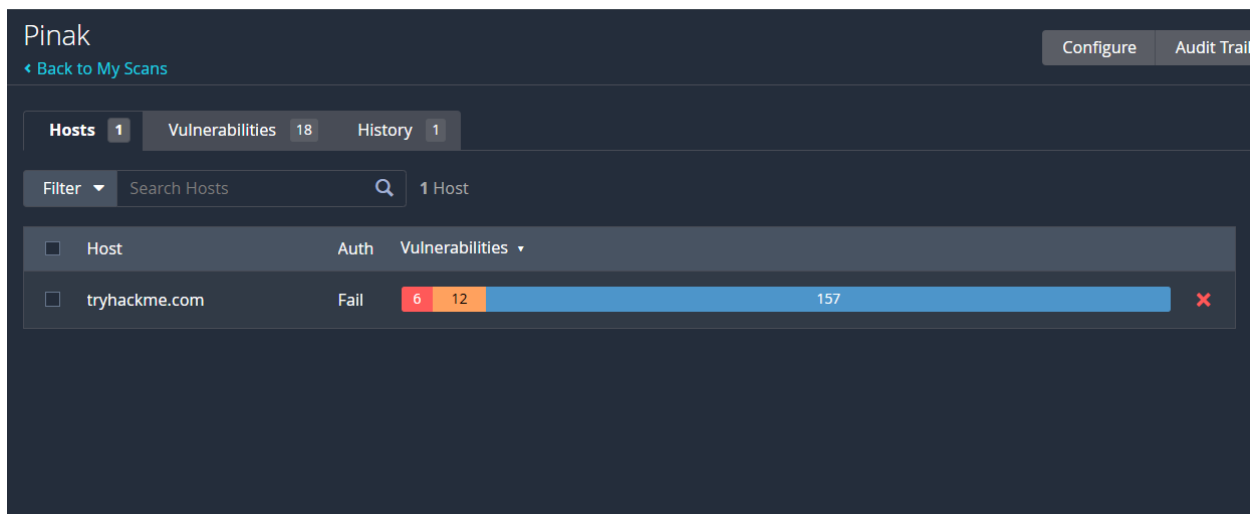
☐	Sev	CVSS	VPR	EPSS	Name	Family	Count	
☐	MIXED	...	...	...	SSL (Multiple Issues)	General	24	
☐	MIXED	...	...	...	TLS (Multiple Issues)	Service detection	24	
☐	INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	32	
☐	INFO	...	...	...	Web Server (Multiple Is...	Web Servers	19	
☐	INFO	...	...	...	TLS (Multiple Issues)	General	18	
☐	INFO				Service Detection	Service detection	19	
☐	INFO				Nessus SYN scanner	Port scanners	13	
☐	INFO				OpenSSL Detection	Service detection	6	
☐	INFO				Remote Services Not Using ...	General	6	
☐	INFO				TLS NPN Supported Protocol...	Misc.	6	
					Common Platform Enumer...	General	1	

my-scans

Common Platform Enumer...

General



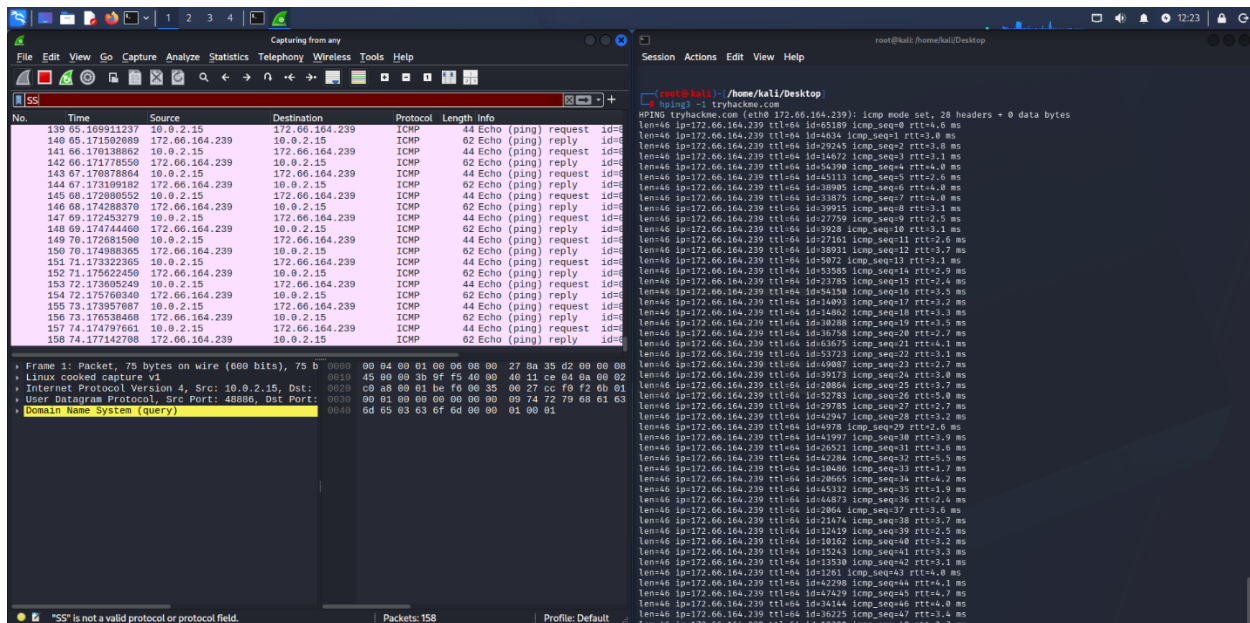


Here we can see that the total vulnerability number is 175 where Falls Positive are 89% which is 157 in number. On Medium & High there are 12 & 6 vulnerabilities. And also we have found no Critical Risk on this site. (tryhackme.com).

Scenario: 3

Executing simulated packet-crafting attack using “hping3 -l tryhackme.com -p” on “tryhackme.com”

Screenshot of Packet Crafting without the Open Port:



Screenshot of Searching Open Port using “Nmap tryhackme.com”

```
(root@kali)-[/home/kali/Desktop]
# nmap tryhackme.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-05 12:29 -0400
Nmap scan report for tryhackme.com (172.66.164.239)
Host is up (0.0013s latency).
Other addresses for tryhackme.com (not scanned): 104.20.29.66 2606:4700:10::6814:1d42 2606:4700:10::ac42:a4ef
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp   open  http-proxy
8443/tcp   open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 5.21 seconds
```

Executing simulated packet-crafting attack using “hping3 -1 tryhackme.com -p 80” on “tryhackme.com”.Using 80 because this is an open port.

Screenshot of Packet Crafting without the Open Port:

The screenshot displays a Wireshark interface with a packet capture of hping3 traffic. The packet list shows multiple ICMP Echo (ping) requests and replies. The packet details pane shows the structure of an ICMP Echo request, including the type, code, and checksum. The packet bytes pane shows the raw hex and ASCII data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
162	76.112559100	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
163	76.116445013	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
164	77.113106807	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
165	77.115491497	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
166	78.114906192	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
167	78.116609192	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
168	79.114862178	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
169	79.117783893	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
170	80.116169623	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
171	80.118287678	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
172	81.116752914	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
173	81.119271521	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
174	82.117259818	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
175	82.120396619	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
176	83.11719732	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
177	83.120142630	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
178	84.117866419	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
179	84.120236647	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...
180	85.118305620	10.0.2.15	104.20.29.66	ICMP	44	Echo (ping) request id=...
181	85.120536666	104.20.29.66	10.0.2.15	ICMP	62	Echo (ping) reply id=...

Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface eth0

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 104.20.29.66

User Datagram Protocol, Src Port: 40584, Dst Port: 80

Domain Name System (query)

Screenshot of the final result After terminating the Attack:

```
— tryhackme.com hping statistic —  
165 packets transmitted, 164 packets received, 1% packet loss  
round-trip min/avg/max = 2.0/3.3/5.3 ms  
  
[ (root@kali) - [/home/kali/Desktop]  
#  
  
[ (root@kali) - [/home/kali/Desktop]  
#
```

SUMMARY:

The system is generally secure and does not contain any critical vulnerabilities. However, several medium and high-level risks, along with multiple exposed services, could still be targeted by attackers. Regular security monitoring, updates, and proper configuration management are recommended to reduce potential risks and maintain system security.